Auditing constrained geospatial allocation under noisy annual land-cover labels

A Geospatial Kernel benchmark in Abu Dhabi

Author: Ning Zhou
Affiliation: Beijing Freedo Technology Co., Ltd.
Corresponding author: Ning Zhou

This study is not an official Abu Dhabi planning forecast. Correspondence: Ning Zhou, Beijing Freedo Technology Co., Ltd.

Abstract

Urban land-change models are often evaluated against annual labels whose reliability is not spatially uniform. We present an auditable protocol for testing constrained spatial allocation under that uncertainty, using Geospatial Kernel, the algorithmic core of a Geospatial World Model, as the focal execution layer. The Abu Dhabi public-data benchmark uses annual 100-m land-cover products from 2017–2024, three computational seeds, destination-correct multi-class change Figure of Merit (FoM), spatial-block bootstrap contrasts and explicit state–action–constraint traces. The regenerated strict FoM was 0.1950 for Geospatial Kernel, 0.1256 for the FLUS-style ANN–CA console and 0.1616 for GeoFM-LDN in 2023; after recursive two-step rollout in 2024 the values were 0.2520, 0.1782 and 0.2708, respectively. On the high-confidence label subset, all three models had FoM 0.000 in 2023 and the 2024 maximum was 0.0188, showing that full-grid agreement is strongly conditioned by label noise. The planning track is therefore reported as scenario stress testing, not official forecasting. The released public bundle, source, checkpoints, rasters, vector footprints and audit manifests support locked-environment reproduction, while cross-platform Kernel reruns differ by about 1% of cells and approximately 0.001 FoM. The protocol's principal contribution is an inspectable proposal–projection–writeback boundary for separating model behaviour from action and constraint semantics.

Highlights

- Geospatial Kernel separates transition proposals from constrained admission.
- Typed state–action–constraint traces make recursive updates auditable.
- Spatial-block uncertainty and paired model contrasts are reported explicitly.
- High-confidence labels reveal near-zero measured change skill.
- Planning outputs are scenario stress tests, not statutory Abu Dhabi forecasts.

Keywords

Geospatial World Model; land-cover change; constrained allocation; cellular automata; Abu Dhabi; scenario planning

Introduction

Land-cover change is a spatial process. Development demand is expressed as a quantity, but its consequences depend on where each new cell is placed relative to roads, existing built areas, ecological features and other land-cover classes. This coupling matters for urban planning because two maps with the same total built area can imply very different infrastructure requirements and environmental pressures. Cellular-automata and suitability-based models have long provided a way to distribute land demand across a map, including CLUE-S, SLEUTH, Dinamica EGO, FLUS, local-attribute urban CA models and more recent patch-generating or deep-learning CA models (Verburg et al., 2002; Silva and Clarke, 2002; Soares-Filho et al., 2002; Lau and Kam, 2005; Liu et al., 2017; Liang et al., 2021). Global Earth-observation products now make it possible to construct annual, spatially aligned inputs for cities that lack a harmonized local data archive (Brown et al., 2022; Gorelick et al., 2017).

Validation studies have also shown that agreement in unchanged cells can mask poor allocation of actual change. Quantity and allocation disagreement, destination-specific transition scores and explicit calibration/validation protocols are therefore preferable to a single overall-accuracy number (Pontius et al., 2008; Pontius and Millones, 2011; van Vliet et al., 2011, 2016). This distinction is central here because the benchmark uses noisy annual labels and tests both one-step allocation and recursive rollout.

The main technical difficulty is not producing another categorical raster. It is keeping the semantics of a planning action separate from the learned transition tendency. A model may predict that a cell is likely to become built, yet the resulting map can still violate a water exclusion, miss the requested class totals or become inconsistent when the next year is simulated from the model's own output. These failure modes are particularly important when the model is used to compare scenarios rather than to reproduce one observed transition. Latent-dynamics approaches such as GeoFM-LDN can represent continuous geospatial state and condition transitions on demand, whereas traditional ANN-plus-cellular-automata systems combine suitability with explicit neighbourhood evolution. Neither design, by itself, guarantees that action semantics, hard constraints, state persistence and evidence provenance are visible at the same execution boundary.

We address this systems problem by studying Geospatial Kernel, the core transition layer of a Geospatial World Model (GWM). The Kernel represents one simulation step as a typed state, a typed action, a transition proposal, a constraint projection and a state writeback. The learned proposal remains domain-specific; the runtime contract is domain-neutral. For Abu Dhabi land-cover planning, the proposal is a six-class probability cube, and the projection allocates only mutable cells until the action's feasible class totals are reached. This separation makes the planning operation inspectable: a reviewer can distinguish what the transition model preferred from what the constraints admitted.

We evaluate this design in a controlled public-data benchmark rather than treating the benchmark as an official data release. All candidates consume the same 100-m grid, common valid mask, annual class-count actions, hard exclusions and evaluator. The historical track uses observed target counts to isolate spatial allocation skill from demand error. The planning track starts from the observed 2024 state and applies three planner-supplied actions: moderate growth, green-priority growth and high outward growth. Abu Dhabi Plan 2030 provides an institutional planning context for discussing urban structure, but the numerical actions in this benchmark are not calibrated to that plan or to an official population forecast (Abu Dhabi Urban Planning Council, 2007). We ask four questions: whether the Kernel execution contract is auditable, whether its allocation skill is competitive, whether it produces feasible and spatially distinct allocations and which conclusions survive label and parameter sensitivity.

The paper makes three bounded contributions. First, it gives a concrete state–action–constraint formulation for using a geospatial kernel as an algorithmic core rather than as an opaque end-to-end predictor. Second, it defines an evaluation protocol that treats high-confidence subset failure as a primary result under noisy annual labels, rather than as a minor diagnostic. Third, it turns future maps into explicit stress-test artefacts, including cumulative rasters and vectorized transition footprints, while preserving the distinction between public-data land cover and authoritative legal land use.

Methods

Study boundary, grid and temporal protocol

The study boundary is the polygon returned for OpenStreetMap relation R4479763, representing Abu Dhabi city. All rasters were aligned to a canonical 100-m grid in EPSG:32640 with 475 columns and 360 rows. A cell was included when its centre fell within the city polygon and when the common public-data mask indicated valid coverage. The benchmark profile records the source geometry, raster transform, dimensions and SHA-256 hashes. The effective evaluation mask contains 79,726 cells.

The observed sequence contains annual states for 2017–2024. Models were fit using the four transitions 2017→2018, 2018→2019, 2019→2020 and 2020→2021. A 2021→2022 transition was used for validation during model development. The historical test starts from 2022 and evaluates 2023 and 2024 in an open-loop rollout without observed-state writeback. The planning test starts from the observed 2024 state and recursively generates 2025–2031 under each scenario action.

Public data and class semantics

Dynamic World V1 provides annual scene-argmax labels and mean top-class probabilities (Brown et al., 2022). The nine Dynamic World labels were cross-walked to six canonical classes: water (source 0), woody vegetation (1), low vegetation (2, 4 and 5), wetland (3), built (6) and bare (7). Source class 8 was excluded. The resulting label is explicitly treated as land cover. No residential, commercial or industrial category is inferred from the raster labels.

Annual 64-dimensional AlphaEarth embeddings were spatially averaged to the canonical grid and locally L2-normalized. Monthly VIIRS night-time-light radiance was averaged by year. Copernicus DEM supplied elevation and a finite-difference slope derivative. OpenStreetMap road geometries were rasterized into distance-to-road and distance-to-major-road surfaces. ESA WorldCover 2021 supplied public water and wetland/mangrove constraint proxies (Zanaga et al., 2022). All dynamic and static layers were checked for exact CRS, transform, width and height agreement.

Geospatial Kernel state, action and transition proposal

Let $S_t(i)$ denote the six-class state at valid cell i, A_t the action for the interval $t \to t+1$, and X_t the aligned driver layers. The Kernel separates a proposal from a projection:

$$Q_t(i, c) = p_\theta(c \mid x_i(S_t, X_t)), \qquad S_{t+1} = \Pi_{\mathcal{C}, A_t}(Q_t, S_t).$$

The feature vector x_i contains 25 values: six one-hot current-class indicators; twelve neighbourhood proportions, comprising 3×3 and 7×7 windows for each of the six classes; and seven continuous variables, namely normalized row and column coordinates, clipped elevation, clipped slope, log-transformed VIIRS radiance, log distance to roads and log distance to major roads. A histogram gradient-boosting classifier estimates the six-class probability vector. It uses learning rate 0.08, 120 iterations, at most 31 leaf nodes, minimum leaf size 40 and L2 regularization 1.0.

Training samples combine all changed valid cells and a 20% random sample of stable valid cells in each fit transition. The sample weight is the product of a change emphasis, $1 + 5\mathbf{1}(S_t \neq S_{t+1})$, and a confidence factor, $0.5 + \min(r_t, r_{t+1})$, where r is the Dynamic World mean top probability. The target years 2023 and 2024 are not used during fitting or model selection. Across the three seeds, each fitted model used 71,892–72,202 pixel rows and all six classes.

Constraint projection and state writeback

For a mutable source cell i of class s and target class c, the base allocation score is

$$q_{i,s\to c} = \log p_\theta(c \mid x_i) - \log p_\theta(s \mid x_i) + \lambda\rho_{i,c}^{(7)},$$

where $\rho_{i,c}^{(7)}$ is the 7×7 fraction of neighbouring cells currently in class c, and $\lambda = 0.35$. The algorithm computes class deficits and excesses from the action's feasible target counts, ranks all admissible source–target candidates by q, and changes cells in descending order until all deficits and excesses are zero. Permanent water and the protected wetland/ecological mask are held fixed. If the counts cannot be satisfied without changing a hard-exclusion cell, the step fails closed rather than silently violating the action.

The projected raster is written as the next `KernelState`. Every step records a state reference, action evidence, model version, parameter reference, projection status and diagnostic counts. The runtime checks domain identity, source-time consistency and action time advancement. This execution contract is the Geospatial Kernel's algorithmic contribution; the Abu Dhabi adapter supplies the land-cover-specific features, learner and constraint masks.

Baselines and common evaluator

The baseline was run through a FLUS-style ANN–CA console using seven continuous drivers: normalized row and column coordinates, elevation, slope, VIIRS radiance, distance to roads and distance to major roads. Its ANN suitability surface was passed to a cellular-automata allocation stage with the common class totals and public hard mask. The archived run used one 2021 training year, eight ANN hidden neurons, a 3×3 CA neighbourhood, neighbourhood strength 1.0, acceleration factor 0.1 and 1,000 iterations. Its frozen transition matrix protected water and wetland but did not prohibit built-to-non-built transitions. The supplied Mach-O executable cannot be traced to a reproducible upstream release or source commit, so we label this control "FLUS-style ANN–CA console (untraceable build)" and do not claim equivalence to the Liu et al. (2017) implementation. A matched 25-feature input attempt terminated with a segmentation fault in the supplied binary before producing a valid probability surface; it is retained as a documented build limitation, not as a completed matched-input comparison.

GeoFM-LDN (Geospatial Foundation-Model Latent Dynamics Network; the repository's former `paper58` path identifier) was implemented as a demand-conditioned residual latent-dynamics network on 64-channel AlphaEarth patches. Three 128-channel convolutions use dilation rates 1, 2 and 4, followed by a 1×1 projection to a 64-dimensional latent state. A 12-dimensional action vector (origin and target counts for six classes) is encoded by a two-layer MLP and broadcast across the patch before concatenation with the embedding. Group normalization and GELU activations are used in the residual blocks. A scikit-learn logistic-regression decoder maps the predicted latent embedding to six semantic classes. Training uses 64×64 patches, batch size 2, eight epochs, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}; the loss is the weighted sum of cosine embedding loss, 0.5 semantic cross-entropy, 0.1 demand-consistency loss and 0.2 hard-constraint consistency loss. The released checkpoints correspond to seeds 31, 47 and 73 and were trained in August 2026. The default reproduction loads these fixed checkpoints rather than retraining; the runner exposes retraining separately. During allocation, GeoFM-LDN directly calls the Geospatial Kernel `allocate_action` routine, so both methods share the same constrained projection and differ primarily in their proposal model. The comparison is therefore a proposal/pipeline comparison, not a comparison of two independent projection architectures. The name describes this benchmark implementation and does not attribute an external publication or performance claim.

The common evaluator computes actual class counts, normalized demand total variation, hard-mask violations, strict multi-class change FoM, binary change FoM, change F1, overall accuracy and macro-F1 for historical runs. The strict FoM counts a hit only when the predicted destination class equals the observed destination class; wrong destination changes are retained in the denominator. It also generates persistence and minimum-change zero controls, spatial-block bootstrap intervals and paired model-difference intervals for strict FoM, overall accuracy and macro-F1. For planning runs it additionally computes ecological conversion, new-built neighbourhood fraction, new-built component counts, all-built component counts, a 500-m leapfrog rate, distances to roads and prior built cells, infrastructure proxy distance and built retirement. Pareto membership is calculated within each scenario over four release objectives: major-road distance, prior-built distance, new-built component density and ecological conversion rate. Vegetation gain is a scenario input and remains descriptive, not an objective.

Scenario actions and uncertainty

The moderate-growth, green-priority-growth and high-outward-growth actions are stored in the tracked `planning_scenarios_public_2025_2031.json` manifest; the legacy `compact` path identifier is retained only for artifact compatibility. Each action supplies feasible class totals for 2025–2031. The 2031 totals are an explicit extension of the previous annual difference and are not derived from an official population, housing or infrastructure forecast. Future exogenous rasters are held at their 2024 values. The green-priority action is not a water-budget model. Each candidate is run at seeds 31, 47 and 73, and planning tables report arithmetic means and population standard deviations. No p-values are reported because the three seeds describe computational variability, not independent field samples. The primary Pareto comparison is within each scenario, contrasting the three models under the same action; a global nine-candidate frontier is retained only as lineage. Sensitivity variants remove or add objective dimensions on the same released rasters. The objective set evolved through four review-driven revisions and was not preregistered: v1 mixed feasibility, action and outcome terms; v2 was not released; v3 used structurally problematic outcomes; and v4 still included scenario-supplied vegetation gain and all-built fragmentation. We therefore report v5 as a release objective set rather than a prespecified planning preference, and treat its sensitivity results as robustness diagnostics.

Change polygons and reproducibility

For each model–scenario pair, annual and cumulative differences from the 2024 raster are extracted as changed 100-m cells and dissolved into GeoPackage layers. The public delivery manifest covers 45 ensemble rasters for 2027–2031 and nine vector packages. The audit script reports `INCOMPLETE_INPUTS` or `FAIL` when source rasters, reports or vectors are absent; with the released public bundle, the input audit passed all alignment gates (with a Dynamic World label-noise warning) and the output audit passed 276 historical and planning predictions with zero failures. The vector layers preserve raster-cell geometry and should not be interpreted as cadastral parcels.

Results

A common benchmark separates allocation skill from demand assumptions

The benchmark covers the Abu Dhabi city polygon represented by OpenStreetMap relation R4479763. The canonical grid is a 475 $\times$ 360 lattice in EPSG:32640 at 100-m resolution. After the common coverage and alignment checks, 79,726 cells were used for scoring. Each cell represents 1 ha. Annual Dynamic World observations from 2017 to 2024 were harmonized to six land-cover classes: water, woody vegetation, low vegetation, wetland, built and bare. AlphaEarth annual embeddings, VIIRS night-time lights, Copernicus DEM derivatives and OpenStreetMap road distances supplied transition drivers. ESA WorldCover and public OpenStreetMap geometries supplied water, wetland and infrastructure exclusion proxies.

The comparison shared the canonical grid, origin states, target actions, hard masks and evaluator, but the public-data run was not information-balanced. The FLUS-style control used seven continuous drivers in its external ANN suitability model, whereas Geospatial Kernel used 25 features including current-class indicators and neighbourhood proportions; GeoFM-LDN used AlphaEarth latent embeddings and the Kernel allocator. We therefore report an unmatched-input pipeline comparison, not evidence that one learner is intrinsically superior. The archived FLUS console is a Mach-O arm64 executable whose original source/version cannot be independently rebuilt from this release, so its results are bounded to the supplied Apple-Silicon binary and are not asserted to be bitwise equivalent to an upstream build. All models used seeds 31, 47 and 73.

The execution trace makes the distinction between proposal and admission explicit. A Kernel step records the source state, action, probability-cube proposal, projected state, model identity, input evidence and next-state reference. The runtime checks that the action advances the source time and that the projected raster becomes the next state. This is the operational meaning of the Kernel in this study: not a claim that all GWM domains share the same learner, but a claim that they can share an auditable execution contract. The shared inputs and execution boundary are summarized in Fig. 1.

Historical and planning results

The historical track uses observed target class totals to isolate spatial allocation from demand estimation. The primary metric is strict multi-class change FoM: a hit requires the predicted destination class to equal the observed destination class, while wrong destination changes remain in the denominator (Pontius et al., 2008). Overall accuracy, macro-F1, change F1 and binary FoM are secondary diagnostics. Uncertainty is estimated with an 8×8-pixel spatial-block bootstrap, and model contrasts are computed from the same resampled blocks for each frozen seed. Persistence and random minimum-change allocation are explicit zero-model controls.

The planning track starts from the observed 2024 state and applies three synthetic class-count actions through 2031. The release objective set is version 5. It uses mean distance of new built cells to major roads, mean distance to pre-existing built cells, connected components formed by newly built cells per 1,000 valid cells and ecological conversion rate. All four are public-data proxies. Vegetation gain is excluded because it is supplied directly by the scenario action, and all-built component density is reported only as a diagnostic because it can be changed by built-cell retirement. No weighted composite score is used. Because the objective set changed across the four review rounds, we disclose the lineage: v1 included seven access, ecological, neighbourhood and gain terms; v2 proposed roads, prior-built distance, fragmentation and leapfrog rate but was not run; v3 used ecological conversion, roads, leapfrog rate and built retirement; and v4 combined roads, prior-built distance, all-built fragmentation and vegetation gain. The current release replaces v4 after the review identified scenario-input leakage and retirement confounding. Sensitivity sets are recomputed on the same rasters and reported as robustness analyses, not as additional preregistered claims. These are measurable public-data proxies rather than claims about welfare, equity or statutory zoning.

The recovered public-data bundle supports a complete re-scoring of the current historical rasters and a re-compilation of the 2025–2031 planning rasters. The strict multi-class transition FoM (mean across three seeds) was 0.1256, 0.1950 and 0.1616 for the FLUS-style console, Geospatial Kernel and GeoFM-LDN in 2023, and 0.1782, 0.2520 and 0.2708 in 2024, respectively. Under this unmatched public-data pipeline, Geospatial Kernel had the largest 2023 value, whereas GeoFM-LDN had the largest 2024 value after recursive two-step rollout; no single model dominated both horizons. These rankings are conditional on the different inputs, learners and shared evaluation projection.

Table 1 | Historical allocation scores. Values are means across the three frozen computational seeds; the strict multi-class change FoM is the primary metric and the remaining columns are diagnostics. Persistence and random minimum-change are explicit zero-model controls rather than learned models.

Target year	FLUS-style ANN–CA	Geospatial Kernel	GeoFM-LDN	Persistence	Random minimum-change
2023 (1-step)	0.1256	**0.1950**	0.1616	0.0000	0.0253
2024 (2-step open-loop)	0.1782	0.2520	**0.2708**	0.0000	0.0605

Spatial-block bootstrap model contrasts support these differences. For 2023, Kernel minus the FLUS-style control was 0.0692 [0.0546, 0.0859] and GeoFM-LDN minus Kernel was -0.0333 [-0.0478, -0.0201]. For 2024, GeoFM-LDN minus Kernel was 0.0184 [0.0046, 0.0321]. These are conditional pipeline contrasts on public labels, not independent-sample significance tests. The persistence control had higher overall accuracy than every learned candidate (0.9436 in 2023 and 0.8938 in 2024), illustrating why change-specific metrics are necessary. Figure 2 shows the four historical metrics alongside the persistence and random minimum-change controls; bars use three-seed population standard deviations, whereas the paired spatial-block intervals above are the uncertainty summaries used for model contrasts.

The reliability sensitivity is substantially less favourable. Requiring both the origin and target Dynamic World mean top probability to exceed 0.5 leaves strict FoM at 0.000 for all three models in 2023; in 2024 the means are 0.004 for the FLUS-style console, 0.005 for Geospatial Kernel and 0.019 for GeoFM-LDN. This is a label-quality diagnostic, not an independent validation set. Dynamic World also reports built pixels increasing from 9,359 in 2022 to 15,598 in 2024 (67%), while the median one-year reversion fraction is 0.3611 and the mean fraction of cells below confidence 0.5 is 0.5497. Test-year change counts are approximately twice the training-transition counts. Because the same grid cells recur across years and the Kernel includes coordinate features, a training–test memory effect cannot be excluded; the reported scores should therefore be interpreted as conditional agreement with the public product.

The reference reproduction environment is macOS arm64 with Python 3.11 and scikit-learn 1.9.0. A Windows rerun with the same source and inputs produced approximately 498–592 differing cells per seed in 2023 and 1,005–1,147 in 2024, or about 1% of the valid grid, with strict FoM differences of roughly 0.001. The paired spatial-block intervals above are much wider than this platform effect and preserve the same model ordering. The manifest therefore supports cross-platform verification of declared files, but not a claim of bitwise identity for floating-point learner outputs.

The planning compiler compares the three models within each of the three scenarios. At 2031, the FLUS-style control has new-built component densities of 4.24–4.67 per 1,000 valid cells, the Kernel 2.97–5.62 and GeoFM-LDN 5.08–6.56. The Kernel is closer to major roads and prior built cells, whereas its fragmentation penalty remains visible. Under the release objective set, the within-scenario frontiers contain FLUS-style and Kernel candidates in all three scenarios; the GeoFM-LDN candidate is additionally non-dominated in the ecological-priority scenario. All sensitivity variants retain the FLUS-style/ Kernel two-model structure; GeoFM-LDN remains additionally non-dominated for ecological-priority growth in the two variants that retain ecological conversion, but not when that objective is omitted. These statements are conditional on public proxy layers and synthetic actions, not model superiority claims. The 2031 majority-vote ensemble L1 demand errors were 8, 44 and 42 pixels for the Kernel's moderate, green-priority and high-outward scenarios, respectively; the corresponding GeoFM-LDN errors were 1,580, 1,524 and 1,148 pixels and the FLUS-style errors were 2,504, 2,888 and 2,568 pixels. Seed-level projections meet their feasible class totals, but majority voting can produce a different raster and therefore a non-zero ensemble error. All 276 historical and planning prediction records passed the raster audit. The resulting 2031 ensemble rasters and their dissolved cell-change footprints are shown in Fig. 5; they are scenario-conditioned land-cover maps, not parcel boundaries or statutory zoning maps.

Table 2 | 2031 within-scenario planning objectives. Values are means across three seeds. Distances are metres, component density is connected new-built components per 1,000 valid cells, and ecological conversion is the fraction of new-built cells whose origin class was woody, low vegetation or wetland. A star denotes membership in the primary within-scenario Pareto frontier.

Model	Scenario	Major-road distance	Prior-built distance	New-built components/1,000	Ecological conversion	Frontier
FLUS-style ANN–CA	Moderate	446.1	231.6	4.666	0.0168	*
Geospatial Kernel	Moderate	336.7	117.4	5.619	0.0000	*
GeoFM-LDN	Moderate	475.5	239.8	5.757	0.0000	

Model	Scenario	Major-road distance	Prior-built distance	New-built components/1,000	Ecological conversion	Frontier
FLUS-style ANN–CA	Green-priority	443.7	212.7	4.578	0.0177	*
Geospatial Kernel	Green-priority	319.8	111.3	5.381	0.0000	*
GeoFM-LDN	Green-priority	452.2	222.8	5.080	0.0000	*
FLUS-style ANN–CA	High outward	443.5	255.3	4.240	0.0131	*
Geospatial Kernel	High outward	362.5	141.7	4.754	0.0000	*
GeoFM-LDN	High outward	507.7	264.4	6.556	0.0000	

Sensitivity shows a stable but not fully explained neighbourhood contribution

The base Kernel allocation score adds a 7 × 7 target-class neighbourhood fraction with weight 0.35. The existing sensitivity and mechanism artefacts are retained as diagnostic controls; they do not identify a causal effect because the proposal features and allocator both contain spatial context. The neighbourhood term is therefore interpreted as evidence about the execution mechanism, not as an independently estimated planning preference. Figure 4A shows the proposal controls, while Fig. 4B–D reports runtime and planning controls.

In the post-hoc neighbourhood-weight sensitivity, the mean strict FoM changed from 0.19997 to 0.20042 across weights 0–0.7 in 2023 and from 0.26079 to 0.26034 in 2024. These changes are small relative to the model contrasts and are not interpreted causally. Objective-set sensitivity on the planning rasters retained the FLUS-style and Kernel candidates in all scenarios. GeoFM-LDN was additionally non-dominated only for ecological-priority growth in the variants that retained ecological conversion.

Mechanism controls separate learned proposal from runtime projection

Mechanism controls separate the learned proposal from runtime projection, hard constraints and recursive writeback. They are not causal policy experiments. Deleting the action reduces mean strict FoM by 0.1124 in 2023 and 0.1699 in 2024, whereas deleting the hard constraint increases it by 0.014 and 0.011, respectively. The former is expected because the historical task supplies oracle target counts; it shows that much of the measured skill belongs to the action-conditioned allocator rather than to a demand forecast. The latter is an important warning: allowing changes inside the exclusion mask can improve agreement with noisy labels while violating the planning contract. The controls therefore support an execution-level interpretation but do not prove that either component causes an urban outcome.

Discussion

This study positions Geospatial Kernel as a candidate execution layer for constrained spatial reasoning and, more broadly, proposes an audit protocol for noisy annual labels. The defensible contribution is that a transition proposal and a planning admission rule can be represented and inspected separately. Under the unmatched public-data pipeline, Geospatial Kernel has the largest strict transition FoM at the one-step horizon, while GeoFM-LDN has the largest value after recursive two-step rollout. The high-confidence subset shows near-zero skill for all models in 2023 and at most 0.0188 FoM in 2024, so full-grid rankings should not be interpreted as validated urban-change skill. We therefore treat the planning frontier as conditional on public proxy objectives and synthetic actions.

The evidence remains bounded. The six classes are remote-sensing land cover, not statutory residential, commercial or industrial land use. Dynamic World labels show substantial annual uncertainty and apparent turnover; no independent 2023–2024 change product or observed 2025–2031 labels was available. Public roads, wetland and protected-area layers are proxies, future drivers are held at 2024, and the green-priority action has no water-budget constraint. The FLUS-style control is an unmatched, untraceable Apple-Silicon binary, and the matched-input attempt failed before producing a valid surface. These public-data stress tests therefore do not establish causal planning effects or a validated Abu Dhabi forecast. An authoritative deployment would replace public labels and masks with locally approved land-use, infrastructure, ecological and development-demand records while retaining the same execution contract.

Data availability

The public-data benchmark manifests, protocols, reports and generated delivery artefacts are organized under `benchmarks/abu_dhabi_land_use_v1/` in the accompanying repository. The main evidence files are:

- `comparison_report_current.json` and `comparison_report_current.md` for historical metrics when a data-complete rerun is available;
- `planning_comparison_report_public_2025_2031_current.json` and its Markdown rendering for 2031 planning objectives;
- `planning_scenario_report_public_2025_2031_current.json` for per-seed rollout traces;
- `planning_public_2025_2031_delivery_manifest_current.json` for raster and vector delivery;
- `output_audit_reproducible.json` for the current raster audit, with `output_audit.json` retained only as legacy lineage;
- `data_audit.json`, `protocol.json`, `gee_input_manifest.json` and `osm_input_manifest.json` for data provenance;
- The released bundle includes the aligned 2017–2024 public rasters, historical seed and ensemble prediction rasters, 2025–2031 planning seed and ensemble rasters, nine GeoPackages of dissolved change footprints, the mechanism-ablation report, model checkpoints, and the SHA-256 manifest. Text hashes use canonical LF line endings, and the byte-length check applies the same normalization on text records, so a Windows checkout with `core.autocrlf=true` can pass the manifest gate. The locked macOS arm64 execution environment remains the reference for model outputs; Kernel reruns on other platforms can differ by about 1% of pixels and approximately 0.001 FoM. No private database address, credential or client-only service is included. Authoritative Abu Dhabi land-use and independent change-validation data remain outside this study.

Code availability

The analysis scripts, a vendored Geospatial Kernel runtime snapshot, benchmark protocol, model runners and figure-generation code are maintained in the accompanying repository. Principal entry points are listed separately to keep the manuscript readable:

- `run_geospatial_kernel.py` for the Kernel historical run;
- `data_agent/uwm/geospatial_kernel/runtime.py` for the vendored runtime snapshot;
- `render_nature_figures.py` for publication artwork.
- `reproducibility/reproduce.py` for the end-to-end rerun;
- `reproducibility_check.py` and `audit_outputs.py` for fail-closed integrity checks.

GeoFM-LDN source and the three fixed checkpoints are included under `benchmarks/abu_dhabi_land_use_v1/external/` and `artifacts/predictions/`. The vendored FLUS executable is a macOS arm64 binary; Linux and Windows users can reproduce the Kernel and GeoFM-LDN tracks, but must provide a compatible FLUS build or treat that baseline as unavailable. The repository is intended to be public; an archival DOI and the corresponding-author e-mail should be added to the submission metadata.

Declarations

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Figure legends

Figure 1 | Benchmark and Geospatial Kernel execution contract. Public annual land-cover, embedding, night-light, terrain and road layers are aligned to a common 100-m Abu Dhabi grid. Each model receives the same state, action and public proxy exclusions. The Geospatial Kernel exposes a learned proposal, constraint projection and state writeback. Historical, open-loop and planning tracks use one audit contract. The schematic does not imply statutory planning boundaries.

Figure 2 | Historical allocation skill. Bars show the revised strict multi-class FoM, change F1, overall accuracy and macro-F1 for the 2023 one-step and 2024 two-step open-loop tests, with persistence and random minimum-change controls. Error bars show population standard deviation across the three frozen seeds; paired spatial-block intervals are reported in the machine-readable comparison report.

Figure 3 | Conditional planning objectives and morphology diagnostics. The 2031 outcomes compare the three models under moderate-growth, green-priority- growth and high-outward-growth actions. Panels A–D show the

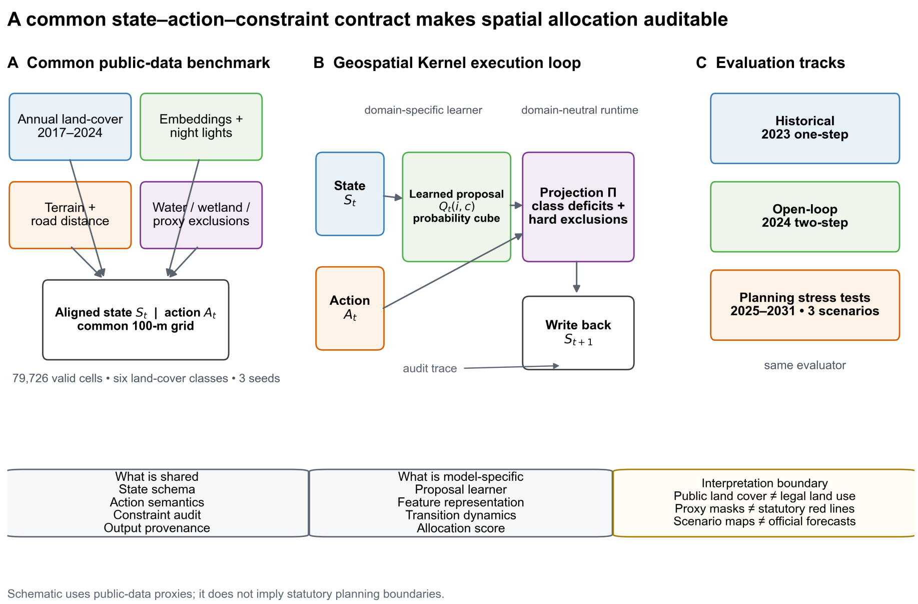

Figure 1: Figure 1: benchmark and execution contract

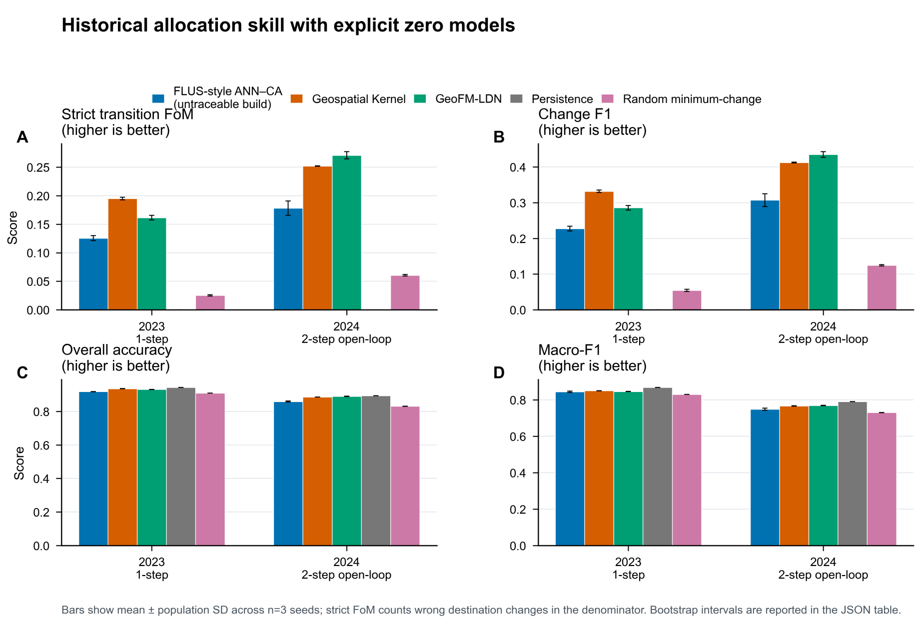

Figure 2: Figure 2: historical allocation skill

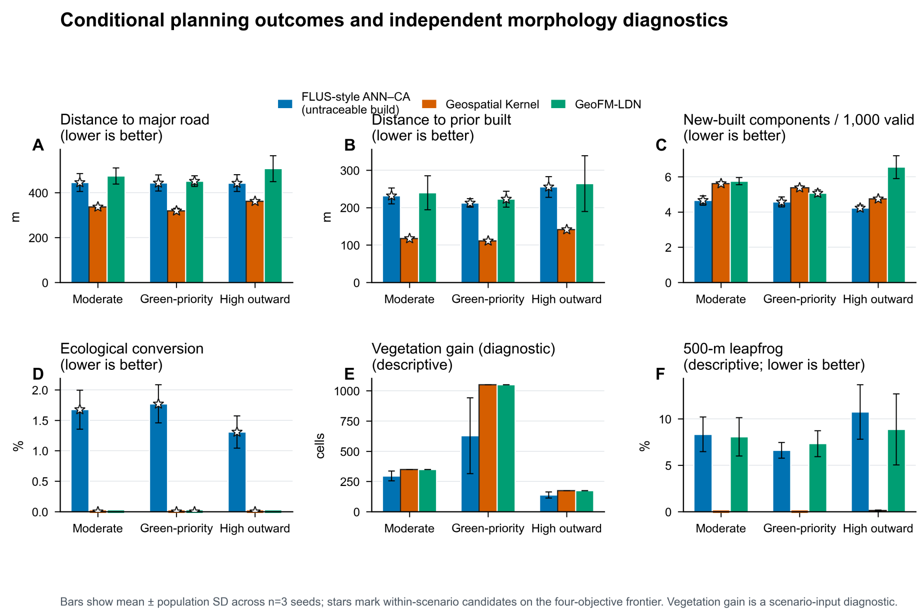

Figure 3: Figure 3: conditional planning objectives

release objectives: distance to major roads, distance to prior built cells, new-built components per 1,000 valid cells and ecological conversion. Panel E shows vegetation gain as a descriptive scenario-input diagnostic; panel F shows the 500-m leapfrog rate. Stars mark non-dominated candidates within each scenario; bars show means ± population SD across three seeds.

Figure 4 | Mechanism controls. Proposal, allocator, action, constraint and writeback controls are compared under the same public-data protocol. Deleting the action lowers historical strict FoM, whereas deleting the hard constraint raises agreement with noisy labels while violating the planning contract. These are diagnostic ablations, not causal policy experiments.

Figure 5 | 2031 spatial footprints. Each panel is a 100-m ensemble raster relative to the observed 2024 state. Overlays identify newly built, newly low-vegetated and retired-built cells; they are raster-cell change footprints, not cadastral parcels.

Supplementary information outline

- **Supplementary Note 1:** Complete data crosswalk, quality metrics and public/proxy status.
- **Supplementary Note 2:** Kernel runtime schemas, audit fields and adapter boundary.
- **Supplementary Table S1:** Full 2025–2031 model–scenario objective

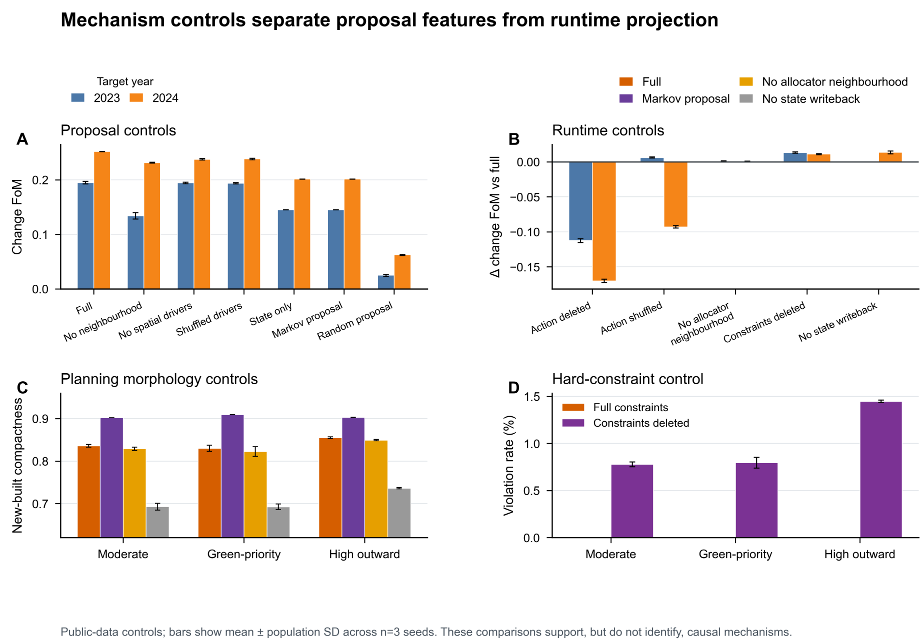

Figure 4: Figure 4: mechanism controls

matrix with per-seed values.

- **Supplementary Table S2:** Neighbourhood-weight sensitivity at 0, 0.175, 0.35 and 0.7.
- **Supplementary Table S3:** Raster and vector delivery audit, hashes and layer counts.
- **Supplementary Figure S1:** Dynamic World confidence and high-confidence change sensitivity.
- **Supplementary Figure S2:** Historical maps and change-error maps for 2024.
- **Supplementary Figure S3:** Driver layers and experiment design in English.

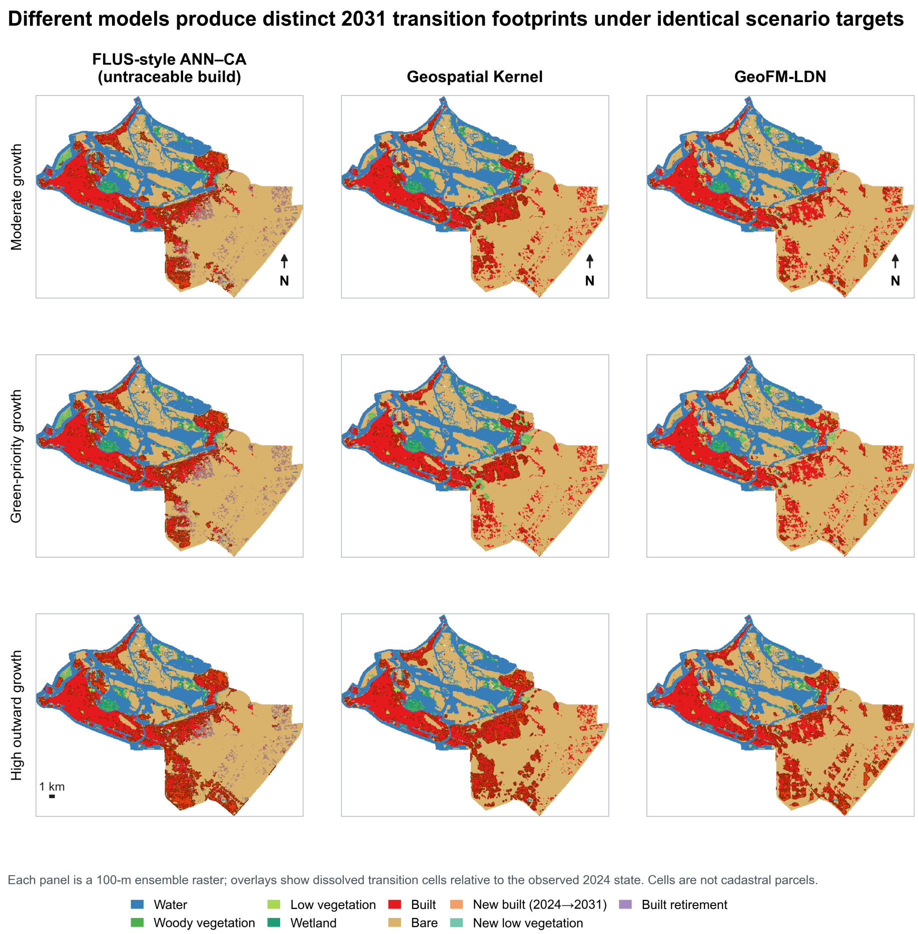

Figure 5: Figure 5: 2031 spatial footprints